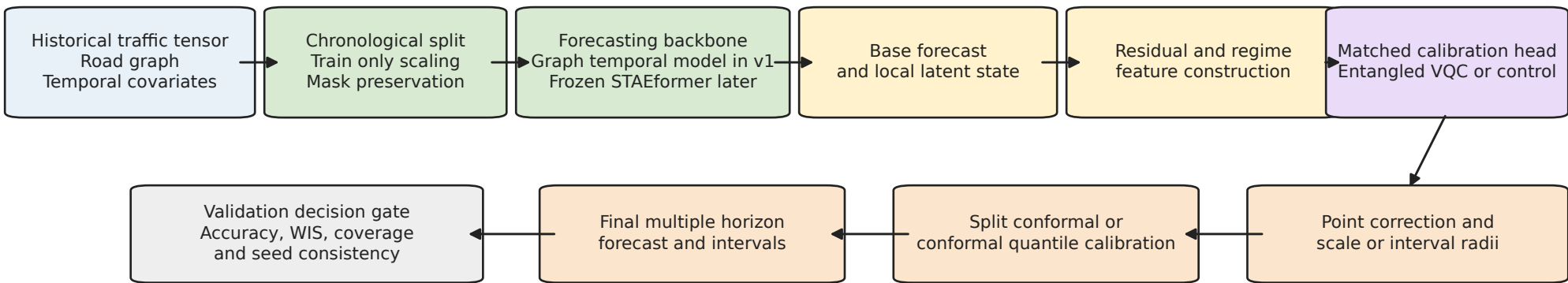
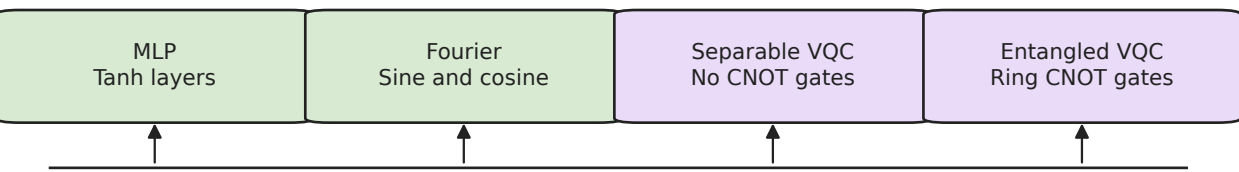


**(a) End to end forecasting and calibration workflow**

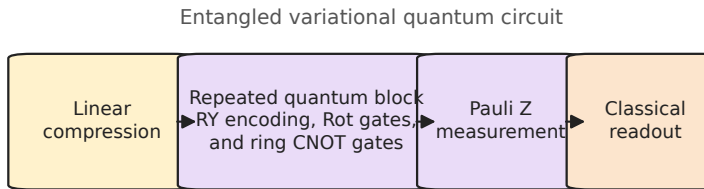


Canonical test evaluation occurs only after the prespecified gate is passed.

**(b) Parameter matched head comparison and entangled circuit**



Identical local features, optimization budget, forecast outputs, and objective



Reference circuit: four qubits, depth two, analytic state vector simulation